



Back from the dead

A tiny marsupial thought to be extinct for 100 years has reappeared in New South Wales, Australia. <http://digit.in/MrsplBack>



NASA Sea level simulator

NASA's sea level simulator lets you bury glaciers in snow, melt ice sheets and trace rising seas as they flood coasts. <http://digit.in/NASASeaSim>

in humans, the effect of behavioural compulsion is the same.

OTHER FACTORS

One of the most common conceptions associated with alcohol and other forms of drug abuse is the eventual degradation and death of brain cells. Research indicates that alcohol abuse eventually reduces the size of the brain, methamphetamine kills cells that produce dopamine and trigger apoptosis, a process where cells self destruct. Needless to say, a brain with fewer cells to do its tasks is less capable of things like judgement, willpower and making the secession to addiction involuntary.

Another factor that influences addiction is genetics. The FosB gene has a splice variant Δ FosB that is agreed to be a critical factor in almost all forms of behavioural and drug addictions. To put it simply, Δ FosB can be described as a "sustained molecular switch" when it comes to developing addiction. And like all genes, it can be inherited – and there's been sufficient research to establish that it has been. For instance, children of alcoholic parents, even when adopted by non-alcoholic parents, have shown greater tendency towards developing an addiction.

PREDICTION

With or without genetics, there are other ways to detect a risk of addiction as well. While the judgement on genes being used to diagnose addiction is still in debate, Henri Begleiter, PhD, Professor of Psychiatry and Neuroscience at the State University

of New York in Brooklyn, New York, has come up with a new method to detect the risk of an addiction through brain waves.

Brain waves are usually measured through an electroencephalograph and printed out as an electroencephalogram (EEG). Anyone experiencing any significant light or sound stimuli shows a peak in the graph of values of about 300-500. For addicts the peak is



Once detected, people can even join treatment programmes and support groups to deal with their addiction in a better way

lower, which means that they cannot distinguish between everyday stimuli and special stimuli, thus processing each one as new, a characteristic called hyperexcitability. This excited state is often dulled down by going after more drugs. This distinct difference in the EEG graphs of addicts and non-addicts can help in early detection of addiction tendencies.

bupropion for nicotine addicts do exactly that.

In addition to that, knowing that there are other factors apart from one's own choice can help individuals and families come to terms with addiction more easily. Taking it one step further, knowing about an inherited tendency towards addiction, people can choose to abstain from substances and addictive behaviour altogether. This does

impact the brain – as people keep away from the people, places and activities that caused their addiction, the brain gradually dissociates the availability of those factors and the eventually induced pleasure from addiction.

BUT...

Even though we have discussed the physiological and neurological details about how addiction works in the previous sections, one of the most important factors behind addiction has been missed out – us. Both drug and behavioural addictions are dependent on us taking the first step or deciding where to draw the line. Irrespective of which gene we've inherited or how our addiction has already affected us, if a choice is made to say a simple 'No' and then stick to it, then it is certainly possible for a mind to win the battle against addiction. 



Brain wave analysis can detect addiction tendencies

THE CURE

The knowledge of the impact neurotransmitters and specific chemicals can have on the functioning of the brain can help create medicine for specific addictions – and it has. For instance, methadone for heroin addicts, naltrexone for alcoholics, and